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Papers 638 (conclusion), 639–646, 647 (commencement)

Arthur Cayley

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[638] ON A q -FORMULA LEADING TO AN EXPRESSION FOR E_1 (CONCLUSION).

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 63–66; continued from preceding pages.]

(Page 26.) so that

$$qX'' = q + 3q^2 + 6q^3 + 10q^4 + \&c.,$$

then

$$X + 8qX = 1 + (9 + 25)q^2 + 49q^3 + 81q^4 + \&c.;$$

so that the right-hand side of (B) is

$$\frac{X + 8qX'}{X} = -1 + 8q \frac{X'}{X}.$$

But (*Fund. Nova*, p. 185, *Ges. Werke*, t. i, p. 237),

$$X = \frac{1 - q^3}{1 - q} \cdot \frac{1 - q^7}{1 - q^5} \cdot \frac{1 - q^{11}}{1 - q^9} \cdots,$$

so that

$$\frac{X'}{X} = \frac{-2q}{1 - q^3} - \frac{4q^5}{1 - q^7} - \frac{6q^9}{1 - q^{11}} - \cdots + \frac{1}{1 - q} + \frac{3q^5}{1 - q^9} + \frac{5q^9}{1 - q^{13}} + \cdots.$$

And the equation (B) intended to be proved thus becomes

$$\begin{aligned} & 1 + 8 \left\{ \frac{-q}{1 + q} + \frac{2q^2}{1 + q^2} - \frac{3q^3}{1 + q^3} + \cdots \right\} \\ & - 16 \left\{ \frac{-q}{1 - q^3} + \frac{2q^2}{1 - q^7} - \frac{3q^3}{1 - q^{11}} + \cdots \right\} \\ & = 1 + 8q \left\{ \frac{-2q}{1 - q^3} - \frac{4q^5}{1 - q^7} - \frac{6q^9}{1 - q^{11}} - \cdots \right. \\ & \quad \left. + \frac{1}{1 - q} + \frac{3q^5}{1 - q^9} + \frac{5q^9}{1 - q^{13}} + \cdots \right\}; \end{aligned}$$

viz. omitting the terms unity, dividing by $8q$, and transposing, this is

$$\begin{aligned} & -\frac{1}{1+q} + \frac{2q}{1+q^2} - \frac{3q^2}{1+q^3} + \dots \\ & + \frac{2}{1-q^3} - \frac{4q}{1-q^7} + \frac{6q^2}{1-q^{11}} - \dots \\ & + \frac{2q}{1-q^3} + \frac{4q^5}{1-q^7} + \frac{6q^9}{1-q^{11}} + \dots \\ & - \frac{1}{1-q} - \frac{3q^5}{1-q^9} - \frac{5q^9}{1-q^{13}} - \dots = 0. \end{aligned}$$

The second and third lines unite together, and the equation becomes

$$\begin{aligned} & -\frac{1}{1+q} + \frac{2q}{1+q^2} - \frac{3q^2}{1+q^3} + \frac{4q^3}{1+q^4} - \dots \\ & + \frac{2}{1-q^3} + \frac{4q}{1+q^7} + \frac{6q^2}{1-q^{11}} + \frac{8q^3}{1+q^{15}} + \dots \\ & - \frac{1}{1-q} - \frac{3q^5}{1-q^9} - \frac{5q^9}{1-q^{13}} - \frac{7q^{13}}{1-q^{17}} - \dots = 0; \end{aligned}$$

(Page 27.) or, collecting and arranging,

$$\begin{aligned} & -\frac{1}{1+q} + \frac{2q}{1+q^2} - \frac{3q^2}{1+q^3} - \frac{4q^3}{1+q^4} - \frac{5q^4}{1+q^5} - \dots \\ & + \frac{1}{1-q} + \frac{3q^2}{1-q^3} + \frac{5q^4}{1-q^5} + \dots = 0, \end{aligned}$$

an identity which it is easy to verify to any number of terms. But to prove it directly, we have only to add the pairs of terms in the alternate columns; calling the left-hand side Fq , we thus obtain

$$Fq = 2q \left\{ -\frac{1}{1+q^2} - \frac{2q^2}{1+q^4} - \frac{3q^4}{1+q^6} - \dots + \frac{1}{1-q^2} + \frac{3q^4}{1-q^6} + \dots \right\};$$

viz. this equation is $Fq = 2qF(q^2)$; and thence

$$Fq = 2^n q^{1+2+4+\dots} F(q^{2^n}) = \&c.;$$

we thus have $Fq = 0$.

The equation (B), or, what is the same thing, the equation (A) is thus proved. Reverting to the equation (A), we have

$$(1 + 2q + 2q^4 + \dots)^4 = \frac{4K^2}{\pi^2}.$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. I, p. 239),

$$\left(\frac{q}{1-q^2} + \frac{2q^2}{1-q^4} + \dots \right) = \frac{K^2}{2\pi^2} \left(1 - \frac{E_1}{K} \right),$$

(ib., p. 135; ib., p. 189),

if $q = e^{-\frac{\pi K'}{K}}$, and K , E_1 are the complete functions $F_1 k$, $E_1 k$.

The left-hand side of the equation is thus

$$\frac{4K^2}{\pi^2} - \frac{8K^2}{\pi^2} \left(1 - \frac{E_1}{K} \right) = \frac{4K^2}{\pi^2} \left(-1 + \frac{2E_1}{K} \right),$$

and we have

$$\left(-1 + \frac{2E_1}{K}\right) = \frac{\pi^2}{4K^2} \cdot \frac{1 - 9q^2 - 25q^6 + 40q^4 + 81q^{10} - \dots}{1 - q^2 - q^4 + q^6 + q^{14} - \dots},$$

which is a new expression for E_1 as a q -function. The expression on the right-hand side presents itself, Clebsch, *Theorie der Elasticität* (Leipzig, 1862), p. 162, and must have been obtained by him as a value for $\left(-1 + \frac{2E_1}{K}\right)$; but there is no statement that this is so, nor anything to show how this form of q -function was arrived at. Mr Todhunter called my attention to the passage in Clebsch.

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639.

AN ELEMENTARY CONSTRUCTION IN OPTICS.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 81, 82.]

(Page 28.) CONSIDER two lines meeting at a point P , and a point A ; through A , drawn at right angles to AP , a line meeting the two lines in the points U, V respectively; and through the same point A draw any other line meeting the two lines in the points U', V' respectively; also let the points u', v' be the feet of the perpendiculars let fall from U', V' respectively on the line UV ; then we have

$$\frac{1}{Au'} + \frac{1}{Av'} = \frac{1}{AU} + \frac{1}{AV}.$$

The theorem can be proved at once without any difficulty. It answers to the optical construction, according to which, if UPV represents the path of a ray through a convex lens AP , then the thin pencil, axis $U'P$ and centre U' , converges after refraction to the point V' , where $U'V'$ are in lined with A the centre of the lens; considering as usual the inclinations to the axis as small, we have approximately $AV' = Av'$, $AU' = Au'$, and the theorem is

$$\frac{1}{AU} + \frac{1}{AV'} = \frac{1}{AU} + \frac{1}{AV},$$

if AF is the focal length of the lens.

In the original theorem, the line UV need not be at right angles to AP , but may be any line whatever; the projecting lines $U'u'$ and $V'v'$ must then be parallel to AP , and the theorem remains true.

640.

FURTHER NOTE ON MR MARTIN'S PAPER "ON THE INTEGRALS OF SOME DIFFERENTIALS."

[From the *Messenger of Mathematics*, vol. vi. (1877), p. 82.]

(Page 29.) See paper, Number 639; this further note relates also to a detail.

641.

ON THE FLEXURE OF A SPHERICAL SURFACE.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 88–90.]

(Page 30.) IT is known that an inextensible spherical surface, or to fix the ideas the spherical quadrilateral included between two arcs of meridian and two arcs of parallel, may be bent in such wise as to be part of a surface of revolution, the meridians and parallels of the spherical surface being meridians and parallels of the new surface, and, moreover, the radius of each parallel of the spherical surface being in the new surface altered in the constant ratio k to 1. We have, in fact, on the spherical surface, writing p for the latitude and q for the longitude, and the radius being unity,

$$x = \cos p \cos q,$$

$$y = \cos p \sin q,$$

$$z = \sin p,$$

values which give

$$dx^2 + dy^2 + dz^2 = dp^2 + \cos^2 p dq^2.$$

This last equation is satisfied by the values

$$x = \cos p \sin \frac{q}{k},$$

$$y = \cos p \sin \frac{q}{k},$$

$$z = E(k, p),$$

where $E(k, p) = \int_0^p \sqrt{(1 - k^2 \sin^2 p)} dp$, is the elliptic function of the second kind; or rather, this is so when $k < 1$, but the same notation may be used when $k > 1$. These values give the deformation in question.

The two cases to be considered are $k < 1$, and $k > 1$; we take in each case a spherical quadrilateral $ABCD$ (fig. 1), bounded by AB (an arc of the equator), the arc of parallel CD , and the two arcs of meridian AD and BC . In the first case, there is no limit to the latitude AD , $= BC$, or taking these $= 90^\circ$, we may in place

(Page 31.) of the quadrilateral $ABCD$ consider the directangular triangle ABE ; the new form of this is $A'B'E'$, where the radius OA' , $= k \cdot OA$, $= k$, is less than the original radius

FIG. 1.

[Figure: spherical quadrilateral with deformation. Original.]

unity, but $OE' = E_1 k$, is greater than the same radius unity. The surface has at E' a conical point, the semi-aperture of the cone being $= \tan^{-1} \frac{1}{k'}$, if as usual $k' = \sqrt{(1 - k^2)}$; to verify this, writing for convenience $q = 0$, we have for the meridian section $x = \cos p$, $z = E(k, p)$, and thence $\frac{dx}{dz} = -\frac{\sqrt{(1 - k^2 \sin^2 p)}}{\sin p}$, which for $p = 90^\circ$ gives $\frac{dx}{dz} = -k'$. Observe also that for $p = 0$, $\frac{dx}{dz} = \infty$, viz. the surface of revolution cuts the plane of the equator at right angles.

There is no limit to the arc AB , it may be $= 360^\circ$, viz. we must in this case cut the hemisphere along a meridian to allow of the deformation; or it may exceed 360° , the hemisphere spherical surface being in this case conceived of as wrapping indefinitely over itself, and we may instead of the half lune $EA'B'$, consider the lune included between two meridians extending from pole to pole, and therefore the whole spherical surface, conceived of as wrapping indefinitely over itself; the result is, that this may be deformed into a surface of revolution, which, in its general form,

resembles that obtained by the revolution of an arc less than a semi-circle round its chord; the half-chord being greater, and the versed-sine less than the radius of the original sphere.

If $k > 1$, there is obviously a limit to the latitude $AD, = BC$, of the spherical quadrilateral; viz. this is equal to $\sin^{-1} \frac{1}{k}$. Supposing that in the quadrilateral $ABCD$ (fig. 1) the latitude has this limiting value, then (see fig. 2) the new form is $A'B'C''D''$, where along the bounding arc $C''D''$ the tangent plane is horizontal; viz. as before $\frac{dx}{dz} = -\frac{\sqrt{(1-k^2 \sin^2 p)}}{\sin p}$, $= 0$ for $p = \sin^{-1} \frac{1}{k}$. It is to be observed, that the radii for the parallels $A''B''$ and $C''D''$ are k and $k \cos p$ respectively; the difference of

(Page 32.) these in $k(1 - \cos p)$, which, however great k is, must be less than the arc of meridian $A''D''$, $= p$; substituting for k the value $\frac{1}{\sin p}$, the condition is $\frac{1-\cos p}{\sin p} < p$, viz. this

FIG. 2.

[Figure: bounded surface with deformation; bell-like shape.]

is $\tan \frac{1}{2}p < p$, which is true for every value up to $p = 90^\circ$. But, more than this, we should have

$$k^2(1 - \cos p)^2 + E^2(k, p) < p^2,$$

viz. writing as before $k = \frac{1}{\sin p}$, this is

$$E^2\left(\frac{1}{\sin p}, p\right) < p^2 - \tan^2 \frac{1}{2}p;$$

this must be true, although (relating as it does to a form of E for which k is greater than 1) there might be some difficulty in verifying it.

There is, as in the first case, no limit to the value of AB , viz. this may be $= 360^\circ$, the spherical zone being then cut along a meridian, or it may be greater than 360° ; and, moreover, the spherical quadrilateral may extend south of the equator, but of course so that the limiting south latitude does not extend beyond the foregoing value $\sin^{-1} \frac{1}{k}$; viz. we may have a zone between the latitudes $\pm \sin^{-1} \frac{1}{k}$, which may be a complete zone from longitude 0° to 360° or to any greater value than 360° . The result is, that the zone is deformed into a surface of revolution, which in its general form resembles that obtained by the revolution of a half-circle or half-ellipse about a line parallel to and beyond its bounding diameter, the bounding half-diameter being less, and the greatest radius of rotation greater, than the radius of the original sphere.

642.

ON A DIFFERENTIAL RELATION BETWEEN THE SIDES OF A QUADRANGLE.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 99–101.]

(Page 33.) LET the sides and diagonals YZ, ZX, XY, OX, OY, OZ of a quadrangle be f, g, h, a, b, c , and let the component triangles be denoted as follows:

$$\begin{aligned} A &= \triangle YZO, &= (b, c, f), \\ B &= \triangle Z XO, &= (c, a, g), \\ C &= \triangle X YO, &= (a, b, h), \\ \Omega &= \triangle X Y Z, &= (f, g, h), \end{aligned}$$

[Figure: quadrangle with vertices X, Y, Z , and interior point O ; subtriangles labelled A, B, C with Ω the outer triangle.]

viz. A, B, C, Ω are the triangles whose sides are $(b, c, f), (c, a, g), (a, b, h), (f, g, h)$ respectively, so that $\Omega = A + B + C$. Then we have between (a, b, c, f, g, h) an equation giving rise to a differential relation, which may be written

$$\Omega (Aa da + Bb db + Cc dc) - (BCff^2 + CAgg^2 + ABh dh) = 0.$$

This may be proved geometrically and analytically. First, for the geometrical proof, it is enough to prove that, when a and b alone vary, the relation between the increments is $Aa da + Bb db = 0$; for then a and g alone varying, the relation between

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(Page 34.) the increments will be $\Omega ada - Cgdg = 0$ (as to the negative sign it is clear from the figure that a, g will increase or diminish together); and we thence at once infer the general relation.

$$Aada + Bbdb = 0;$$

or, what is the same thing,

$$ada : -bdb = XOZ : YOZ.$$

The points XYZ remain fixed; but O moves through the infinitesimal arc OO' , centre Z , which may be considered as situate in the right line OM drawn from O at right angles to ZO , and meeting XY produced in the point M . And then, writing for a moment $\angle OXY = X, \angle OYX = Y, \angle OMY = M$, we find at once

$$da = OO' \cos(X + M),$$

$$-db = OO' \cos(Y - M);$$

that is,

$$\frac{da}{db} = \frac{\cos(X + M)}{\cos(Y - M)}, \text{ or } \frac{ada}{bdb} = \frac{a \cos(X + M)}{b \cos(Y - M)}.$$

But drawing $X\alpha, Y\beta$ each of them at right angles to ZO , we have $a \cos(X + M) = X\alpha, b \cos(Y - M) = Y\beta$, and evidently $XOZ : YOZ = X\alpha : Y\beta$; whence the equation is $\frac{ada}{bdb} = \frac{XOZ}{YOZ}$, which is the required relation.

For the analytical proof, it is to be observed that the relation between a, b, c, f, g, h is a quadric relation in the quantities $a^2, b^2, c^2, f^2, g^2, h^2$ respectively; this may be written

$$\left| \begin{array}{cccccc} 1 & a^2 & a^4 & & & \\ 1 & b^2g^2 + c^2h^2 + a^2f^2 - (b^2 + c^2)g^2h^2 - (g^2 + h^2)b^2c^2 & -(b^2 - c^2)(g^2 - h^2) & & & \\ \nu = & + (b^2 - h^2)(c^2 - g^2) & -b^2 - c^2 - g^2 - h^2 & +1 & & \\ f^2 & & & & & \\ f^4 & & & & & 1 \end{array} \right| = 0;$$

say for a moment this is $A + Ba^2 + Ca^4 = 0$, where

$$\begin{aligned} A &= b^2g^4 + b^4g^2 + c^4h^2 + c^2h^4 + f^4(g^2 - h^2)(c^2 - g^2) \\ &\quad - (b^2 + c^2)g^2h^2 - (g^2 + h^2)b^2c^2 + f^2(-b^2 - c^2 - g^2 - h^2) + f^4, \\ B &= -(b^2 - c^2)(g^2 - h^2) + f^2(-b^2 - c^2 - g^2 - h^2) + f^4, \\ C &= 1; \end{aligned}$$

then we have as usual

$$\frac{da}{d \cdot a^2} ada + \frac{da}{d \cdot a^2} bdb + \&c. = 0,$$

(Page 35.) where

$$4 \frac{da}{d \cdot a^2} = Ca^2 + \frac{1}{2}B.$$

But in virtue of $u = 0$, we have

$$(Ca^2 + \frac{1}{2}B)^2 = C(Ca^4 + Ba^2 + A) + \frac{1}{4}(B^2 - 4AC),$$

that is, $\frac{da}{d\alpha^2} = \sqrt{(B^2 - 4AC)}$; and hence $B^2 - 4AC$ is a quartic function of f^2 , which is easily seen to reduce itself to the form

$$f^2 - (g + h)^2 f^2 - (g - h)^2 f^2 - (b + c)^2 f^2 - (b - c)^2;$$

The coefficients of bdb , cdc , &c., are given as expressions of the like form; substituting their values, the differential relation is

$$\sqrt{(f^2 - (g + h)^2 f^2 - (g - h)^2 f^2 - (b + c)^2 f^2 - (b - c)^2)} ada + \&c. = 0,$$

which is, in fact, the foregoing result.

It is right to notice that there are in all 16 linear factors,

$$\begin{array}{llll} f + g + h, & b + c + f, & c + a + g, & a + b + h \text{ say } d, f, g, h, \\ -f + g + h, & -b + c + f, & -c + a + g, & -a + b + h & d', f', g', h', \\ f - g + h, & b - c + f, & c - a + g, & a - b + h & d'', f'', g'', h'', \\ f + g - h, & b + c - f, & c + a - g, & a + b - h & d''', f''', g''', h'''; \end{array}$$

and this being so, the coefficients of ada , bdb , cdc , fdf , gdg , hdh , are

$$\begin{array}{l} \sqrt{(dd'd''d''' \cdot f f' f'' f''')}, \sqrt{(gg'g''g''' \cdot hh'h''h''')}, \\ \sqrt{(dd'd''d''' \cdot gg'g''g''')}, \sqrt{(hh'h''h''' \cdot f f' f'' f''')}, \\ \sqrt{(dd'd''d''' \cdot hh'h''h''')}, \sqrt{(f f' f'' f''' \cdot gg'g''g''')}, \end{array}$$

respectively.

We may imagine the quadrilateral $ZOXY$ composed of the four rods ZO , OX , XY , YZ (lengths c , a , h , f as before) jointed together at the angles, and kept in equilibrium by forces B , G acting along the diagonals $OY(=b)$, $ZX(=a)$ respectively. We have c , a , h , f given constants, and the relation $\phi(a, b, c, f, g, h) = 0$, which connects the six quantities b and g ; by what precedes, the differential relation $\phi_y, dg + \phi_\delta \cdot db = 0$ is equivalent to the foregoing four equations equal to the differential dt of a new variable, the system of the equations being expressed by $\delta\delta b + Gdg = 0$; by virtual velocities we have as the condition of equilibrium $Bdb + Gdg = 0$; hence, eliminating db , dg we have

$$\frac{B}{\Omega Bbb} = \frac{G}{CAgg},$$

or, say

$$\frac{B}{b} = \frac{G}{g} = \frac{1}{\sqrt{\Delta XYO \cdot \Delta ZYO}} : \frac{1}{\sqrt{\Delta XYZ \cdot \Delta ZXO}},$$

viz. the forces, divided by the diagonals along which they act, are proportional to the reciprocals of the products of the two pairs of triangles which stand on these diagonals respectively. The negative sign shows, what is obvious, that the forces must be, one of them a pull, the other a push.

643.

ON A QUARTIC CURVE WITH TWO ODD BRANCHES.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 107, 108.]

(Page 36.) IT is a known theorem that the branches of a plane curve are even or odd; viz. two even branches, or an even and an odd branch (whether of the same curve or of different curves) intersect in an even number of real points; but two odd branches (of the same curve or of different curves) intersect in an odd number of real points*.

In particular, a right line is an odd branch, and hence it meets any even branch of a curve in an even number of real points, and an odd branch in an odd number of real points; or (what is the same thing) an even branch is one which can be drawn throughout) of real points; an odd branch is one which is met by any right line whatever in an odd number of real points.

A curve without nodes has at most one odd branch; for if there were two, one of them being an asymptote (it is by no means certain what has been called a *complete* branch, viz. the partial branches which touch an asymptote at its opposite extremities are considered as parts of one and the same branch, and so in other cases. Thus a quadric curve, whether ellipse, parabola, or hyperbola, is one even branch; a cubic curve, in either one odd branch, or else it is an odd branch and an even branch; and generally a curve of an odd order has always an odd branch of odd branches, and a curve of an even order has always an even number of odd branches.

A curve without nodes has at most one odd branch; for if there were two, they would intersect in a real point, which would be a real node on the curve. In particular, a quartic curve having two odd branches must have a real node; this however may be, as in the instance about to be given, an isolated point of infinity.

A simple instance of a quartic curve with two odd branches is that represented by the equation

$$(x^2 - 1)(y^2 + 1) - 2nxy = 0,$$

* The two branches must be distinct branches; a branch whether odd or even does not of necessity intersect itself (have upon it any real node), but it may intersect itself (have upon it any odd, or an even, number of real points.)

(Page 37.) or, what is the same thing,

$$y = \frac{1}{x^2 - 1} \left[mx \pm \sqrt{\left\{ -(x^2 - 1) \left(x^2 - \frac{1}{n} \right) \right\}} \right],$$

where

$$n^2 + \frac{1}{n^2} = 2 + m^2,$$

or say

$$n^2 = \frac{1}{2} \left[2 + m^2 + m\sqrt{(4 + m^2)} \right],$$

$$\frac{1}{n^2} = \frac{1}{2} \left[2 + m^2 - m\sqrt{(4 + m^2)} \right],$$

so that m being positive $n > 1$, and the curve consists of two real branches included between the lines $x = a$, $x = -\frac{1}{a}$, and the lines $x = -a$, $x = -\frac{1}{a}$ respectively; each of these lines touches the curve in a real point, viz. a having any one of the last-mentioned values, the value of y at the point of contact is $y = \frac{ma}{a^2 - 1}$; and between each pair of lines we have the asymptote $x = +1$ or $x = -1$. Hence the curve has the form shown in the figure, and it is thereby evident, that each branch of the curve is met by

[Figure: quartic curve showing two odd branches with asymptotes $x = \pm 1$ and the nodes.]

any real right line whatever in one real point, or else in three real points. The numerical values in the figure are $a = \frac{2}{3}$, $m = \frac{1}{2}$, whence also $x = a$ or $-\frac{1}{a}$, $y = 1$, and $x = -a$ or $\frac{1}{a}$, $y = -1$.

The curve has two nodes at infinity, viz. writing the equation in the form

$$(x^2 - z^2)(y^2 + z^2) - 2nxyz^2 = 0,$$

that is,

$$x^2y^2 + z^2(x^2 - y^2 - mxy) + z^4 = 0,$$

it appears that the points $(x = 0, z = 0)$, $(z = 0, y = 0)$ are each of them a node. The first of these $(x = 0, z = 0)$ is the real intersection of the two odd branches; the other of them is a conjugate point.

644.

NOTE ON MAGIC SQUARES.

[From the *Messenger of Mathematics*, vol. vi. (1877), p. 168.]

(Page 38.) IN a magic square of any odd order, formed according to the ordinary process, there is a tolerably simple analytical expression for the number which occupies any given compartment; thus taking the square of 21, let the dexter diagonals (N.W. to S.E.) commencing from the N.E. corner compartment, be numbered $1, 2, 3, \dots, 20, 21, 20', \dots, 2', 1'$, the diagonals of course containing these numbers of compartments respectively; and in any diagonal let the compartments reckoning from the top line be numbered $1, 2, 3, \dots$, respectively; then if $D_{\theta, \phi}$ (or $D'_{\theta, \phi}$ as the case may be) denotes the number in the compartment ϕ of the diagonal θ or θ' , we have

$$\begin{aligned} D_{\text{odd}, \phi} &= 20\theta + 10 + \phi, \\ D_{\text{even}, \phi} &= 20\theta + 231 + \phi(-21), \\ D'_{\text{odd}, \phi} &= -22\theta + 430 + \phi, \\ D'_{\text{even}, \phi} &= -22\theta + 231 + \phi(-21), \end{aligned}$$

where in the second and fourth expressions the term -21 is to be retained only if $\phi > \theta$; if $\phi \not> \theta$, it is to be omitted. There would be a like formula for a square of any odd order, and it would be easy to write down the formulae for the general value $2n + 1$; but I have preferred to give them for a specific case.

645.

A SMITH'S PRIZE PAPER, 1877.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 173–182.]

(Page 39.) THE paper was as follows:

1. Show (independently of the theory of roots) how, if x satisfies an equation of the order n , a given rational function of x can in general be expressed as a rational and integral function of the order $n - 1$. State the theorem in a more precise form, so as to make it true universally.
2. Investigate the form of the factors of $1 \pm \sin(2n + 1)x$ considered as a function of $\sin x$; and give the formulae in the two cases, $2n + 1 = 3$ and 5 respectively.

3. Write down the substitutions which do not alter the function $ab + cd$; and explain the constitution of the group.
4. Find in a form adapted for calculation an approximate value for the sum of the middle $2x + 1$ terms of the expansion of $(1 + 1)^n$, n being a large number, and x small in comparison therewith.
Obtain thence a complete and precise statement of the theorem that in a large number of tosses the numbers of heads and tails will probably be nearly equal.
5. A point in space is represented on a given plane by its projections from two fixed points. Show how a problem relating to points, lines, and planes, is thereby reduced to a problem *in plano*; and apply the method to construct the line of intersection of two given planes each passing through three given points.
6. A weight is supported on a tripod of three unequal legs resting on a smooth horizontal plane, their feet connected in pairs by strings of given lengths. Show how to determine the tensions of the several strings.
7. Explain the ordinary configuration of a system of isoparametric lines on a spherical surface; for instance, what is the configuration when there are two points of minimum value, and one point of maximum value, of the parameter?

(Page 40.)

8. Find the attraction of an infinite circular cylinder, of uniform density, on a given exterior or interior point.
9. Determine the number of arbitrary constants contained in the equation of a surface of the order r which passes through the curve of intersection of two given surfaces of the orders m and n respectively.
10. Find, for the several values of p , the number of the conics passing through p given points and touching $5 - p$ given lines; and, in each case, show how to obtain (in point-coordinates or line-coordinates, as may be most simple) the equations of the conics satisfying the conditions in question.
11. Investigate the theory of the linear transformation of a ternary quadric function into itself.
12. Explain the theory of the solution of a partial differential equation, given function of $x, y, z, p, q, r = 0$, viz. the equation involves only the dependent variable u in regard to the independent variables x, y, z respectively.

I propose, not (as in former years) to give complete solutions, but only to notice in more or less detail the leading points in the several questions.

1. The expression is of course required, not only for a given integral function of x , but for a given fractional function. The case where the given function is integral presents no difficulty; when the given function is fractional, the most simple case is where it is $= \frac{1}{x-a}$, supposing the equation to be $f(x) = 0$, here dividing $f(x)$ by $x - a$, we have a quotient $X(x)$ which is a rational and integral function of an order not exceeding $n - 1$, and a remainder $f(a)$ which is a rational and integral function of a . Hence dividing

$$\frac{f(x)}{x - a} = R(x) + \frac{f(a)}{x - a},$$

or, in virtue of the given equation $\frac{f(x)}{x-a} = -R(x)$, viz. we have thus $\frac{1}{x-a}$ in the required form. But if $f(a) = 0$, then we do not obtain such an expression of $\frac{1}{x-a}$. It has to be shown that the like considerations apply to any fractional function of x , and the precise form of the theorem in the case of a rational fractional function of x which by their intersection determine the nodes, have in common the line $(x = 0, y = 0)$, and there is a diminution in the number of their common intersections. I do not pursue the enquiry, but pass to a different question.

(Page 41.) that is, $0 = \frac{\cos(2n+1)x}{\cos x}$, the value in question gives $\sin(2n+1)x = 1$, and therefore $\cos(2n+1)x = 0$; and it does not give $\cos x = 0$; hence every such factor $1 - \frac{\sin x}{\sin a}$ is a double factor, viz.

$$1 - \sin(2n+1)x = (1 \pm \sin x) \prod \left(1 - \frac{\sin x}{\sin a}\right)^2.$$

Or the like result may be obtained by considering instead of $1 - \sin(2n+1)x$, the more general function

$$\sin(2n+1)a \pm \sin(2n+1)x,$$

and finally assuming $a = \frac{1}{2}\pi$.

3. Relates to a theory which is not, but ought to be, treated of in the text books of the University. See Serret's *Algèbre Supérieure*, t. II., Sect. IV.

The substitutions which leave $ab + cd$ of unaltered are

1 1, that is, the letters remain unchanged,

α (ab) , that is, a and b are interchanged,

β (cd) , that is, c and d are interchanged,

γ $(ab)(cd)$, that is, a and b and also c and d are simultaneously interchanged,

δ $(ac)(bd)$, same with a and d , b and c ,

ϵ $(ad)(bc)$, same with a and d , b and c ,

ζ $(acbd)$, that is, we cyclically change a into c , c into b , b into d , and d into a ,

θ $(adbc)$, that is, we cyclically change a into d , d into b , b into c , and c into a ,

viz. we have eight substitutions 1, α , β , γ , δ , ϵ , ζ , θ forming a group; that is, the product of any two of them, in either order, is a substitution of the group (or, what is the same thing, the same as that of the performance thereon of some other substitution of the group); thus we have $\alpha^2 = 1$, $\beta^2 = 1$, $\gamma^2 = 1$, $\alpha\beta = \beta\alpha = \gamma$, &c.; the system of these equations, which verify that the set of substitutions form a group, defines the constitution of the group—thus to take a more simple instance, of 4 may be 1, a , a^2 ($a^3 = 1$) or 1, a , b , ab ($a^2 = 1$, $b^2 = 1$, $ab = ba$).

4. The expression of the general coefficient is

$$= \frac{1 \cdot 2 \dots 2n}{1 \cdot 2 \dots n - x \cdot 1 \cdot 2 \dots n + x},$$

which can be transformed by the well-known formula

$$1 \cdot 2 \dots u = u^{u+\frac{1}{2}} \sqrt{(\pi)} e^{-u},$$

(Page 42.) viz. the coefficient thus becomes

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} \cdot \frac{1}{\left(1 - \frac{x}{n}\right)^{n-x+\frac{1}{2}} \left(1 + \frac{x}{n}\right)^{n+x+\frac{1}{2}}}.$$

Now x is supposed small in comparison with n , and the factors in the denominator have the logarithms

$$(n - x + \tfrac{1}{2}) \log \left(1 - \frac{x}{n}\right) = (n - x + \tfrac{1}{2}) \left(-\frac{x}{n} - \tfrac{1}{2} \frac{x^2}{n^2}\right), \quad = -x + \tfrac{1}{2} \frac{x^2}{n},$$

and

$$(n + x + \tfrac{1}{2}) \log \left(1 + \frac{x}{n}\right) = x + \tfrac{1}{2} \frac{x^2}{n};$$

hence the denominator is $= e^{\frac{x^2}{n}}$, and the final approximate value of the coefficient is

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} e^{-\frac{x^2}{n}}.$$

Hence, converting as usual the sum into a definite integral, we have the sum of the $2x + 1$ coefficients

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} \int_{-x}^x e^{-\frac{x^2}{n}} dx,$$

or, what is the same thing,

$$= \frac{2^{2n}}{\sqrt{(\pi)}} \int_{-\frac{x}{\sqrt{n}}}^{\frac{x}{\sqrt{n}}} e^{-x^2} dx.$$

For the chance that the number of tosses lies between $n + x$ and $n - x$, this has merely to be divided by 2^{2n} ; hence writing $x = kn$, the chance that the number may be between $n(1 + k)$ and $n(1 - k)$ is

$$= \frac{1}{\sqrt{(\pi)}} \int_{-k\sqrt{n}}^{k\sqrt{n}} e^{-x^2} dx,$$

where observe that the integral, taken with the limits $\infty, -\infty$ has the value $\sqrt{(\pi)}$.

Considering k as a given fraction however small, by increasing n we make $k\sqrt{(n)}$ as large as we please, and therefore the integral, as nearly as we please $= \sqrt{(\pi)}$, or the chance as nearly as we please $= 1$; and hence the complete and precise statement of the theorem, viz. by sufficiently increasing the number of tosses, the probability that the deviation from equality shall be any given percentage (as small as we please) of the deviation from equality is unity. The differential equation was considered by the Rev. H. J. Sharpe, who mentioned it to Prof. Stokes.

Further, restoring x in place of kn , the chance of a number between $n + x$ and $n - x$ is

$$= \frac{1}{\sqrt{(\pi)}} \int_{-\frac{x}{\sqrt{n}}}^{\frac{x}{\sqrt{n}}} e^{-x^2} dx,$$

(Page 43.) which when $\frac{x}{\sqrt{(n)}}$ is small is $= \frac{2x}{\sqrt{(n\pi)}}$, (more accurately $\frac{2x+1}{\sqrt{(n\pi)}}$, when x is small); hence, however large x is, the chance of a deviation from equality not exceeding $\pm x$ is as small as we please.

5. The point is represented in the given plane by two points which lie in lines (say O) at a fixed point (say O) of that plane, the line O is the intersection of the given plane by the line which joins the two projecting points.

A plane may be represented in the given plane by two lines, viz. these are the projections of the two lines which it makes with the two planes through the projecting points O .

A plane may be represented in the given plane by two lines, viz. these are the projections of the line in which it joins to projecting points.

6. The solution is represented on the given plane by means of virtual velocities; taking a, b, c for the lengths of the legs, f, g, h for the lengths of the strings, and z for the height of the summit, z is a known function of a, b, c, f, g, h (z is in fact $= \frac{3V}{\Delta}$, where V , the volume of the tetrahedron, is a given function of a, b, c, f, g, h ; and Δ , the area of the base, is a function of a, b, c). Writing then F, G, H for the tensions, and W for the weight, and regarding z, f, g, h as variable, the principle gives

$$Wdz + Fdf + Gdg + Hdh = 0,$$

that is,

$$F, G, H, = -W \frac{dz}{df}, -W \frac{dz}{dg}, -W \frac{dz}{dh},$$

respectively.

7. The ordinary case is when no isoparametric lines has on one side of it larger values, on the other side smaller values, of u , and M the general function of the order $r - m - n$, then we have a value of $L + MV = 0$, where the general number L of the order r . Supposing only that $r > m + n$, this is the general number r where f is the general function of the order r , L that of the order $r - m$, and M that of the order $r - m - n$; the case where $r - m$ or $r - m - n$ is negative giving rise to an exception, the case being that no integration is performed by the equation of the surface or surface, of any maximum, parameter, the case excluded.

(Page 44.) The lines in the neighbourhood of a point of maximum, or of minimum, parameter are ovals surrounding this point; in question, each oval being itself surrounded by the consecutive oval. Supposing that there are two points of maximum and one of minimum, parameter, we have round each of them a series of ovals, until at length an oval belonging to the one of them ceases to unite itself with a series of ovals, until of the points of maximum, parameter, into a single point of maximum, parameter; and the remaining curves on the series are then a series of continually diminishing closed curves, and the figure of eight itself surrounded by a series of continually increasing closed curves.

8. The investigation by means of the Potential presents the difficulty that the Potential of the infinite cylinder has no determinate value, as at once appears from the constructive oval. Supposing that there are two points of maximum, and one of minimum parameter, we have round each of them a series of ovals and oval belonging to the one of them ceases to unite itself with a series of ovals; thus the two ovals of maximum, parameter, into a single point of maximum, parameter; and the remaining curves on the series are then a series of continually diminishing closed curves, and the figure of eight itself surrounded by a series of continually increasing closed curves.

The lines in the neighbourhood of a point of maximum, or of minimum, parameter are ovals surrounding this point in question, each oval being itself surrounded by the consecutive oval. Supposing that there are two points of maximum and one of minimum, parameter, we have round each of them a series of ovals, until at length an oval belonging to the one of them ceases to unite itself, with a series of ovals; thus, the two ovals of the points of maximum, parameter; and the remaining curves on the series are then a series of continually diminishing closed curves, and the figure of eight itself surrounded by a series of continually increasing closed curves.

It is necessary, however, before quitting the question, to consider the case where $\frac{dV}{dx}, \frac{dV}{dy}$ contain only differential coefficients of V , are determinate. But it is safer to work directly with the Attraction; the Attraction of an infinite line acts in the perpendicular plane through the attracted point, and is inversely proportional to the distance; the problem is thus reduced to the plane

problem of a circle of uniform density, force varying as the distance (distance), attracting a point in its own plane. This is precisely similar to the case of a sphere with the ordinary law of attraction, dividing the circle into rings, each ring is a portion of an interior point, and an attraction as if its mass were collected at its centre, for an exterior point, and as $r - m - n$.

10. The numbers are known to be 1, 2, 4, 4, 2, 1, which values are obtained most easily (though not in the way which is theoretically most interesting) by finding for

(Page 45.) the first three cases the equation of the required conic in point-coordinates; and then, by changing these into line-coordinates, we have for points, the equations of any of the required conic in line-coordinates, and the condition of touching a given line gives a quadric for λ ; the number is therefore = 2.

$p = 5$: 1 point. The equation of the conic is

$$(a, b, c, f, g, h, y, z)^2 = 0,$$

and we have 5 linear equations to determine the ratios of the coefficients; the number is therefore = 1.

$p = 4$: 4 points and 1 line. Taking $U = 0$ and $V = 0$, the equations of any of two conics each passing through the four points, the equation of the required conic must be of the form $U + \lambda V = 0$, and the condition of touching it gives a quadric in λ ; the number is therefore = 2.

$p = 3$: 3 points and 2 lines. Taking $U = 0$ and $V = 0$, the conics containing the three conics through the three points, the equations are $\sigma = a, z = c, z = b, z = b$ are the equations of any conic containing these passes through the three points; the conditions of touching the given lines is $y = \frac{ma}{a^2-1}$; and between each pair of conics for $\sqrt{f}, \sqrt{g}, \sqrt{h}$; and therefore each of the eight values of $\sqrt{f}:\sqrt{g}:\sqrt{h}$; that is, the number is = 4.

$p = 2$: 2 points and 3 lines. Taking $z = a, z = b, z = c$, as the equations of the three lines, $z = b, z = c, \xi a, \eta b, \zeta c$, respectively. The equation of the conic in question is taken to be

$$\sqrt{\sqrt{\xi} + \sqrt{\eta} + \sqrt{\zeta}} = 0,$$

where each radical multiplied; viz. this is the equation in line-coordinates and adding we have the line equation in line-coordinates of conics touching the three lines $\xi = 0, \eta = 0, \zeta = 0$, and the four values of $\sqrt{f}:\sqrt{g}:\sqrt{h}$;

$$a\xi + b\eta + g\zeta = ax + by + gz, \quad \text{etc.,}$$

$$b\xi + c\eta + f\zeta = bx + by + fz, \quad -\lambda\zeta,$$

$$g\xi + f\eta + c\zeta = gx + fy + cz, \quad +\lambda\eta.$$

(Page 46.) where λ, μ, ν are arbitrary; viz. multiplying by ξ, η, ζ and adding we have the equation in question. The two conics determine three values of x, y, z and the equation in question then becomes three independent variables x, y, z respectively.

12. The partial differential equation in the form actually equally well have been given here, $f(x, y, z, p, q, r) = 0$, the equation then is $\phi(x, y, z, p, q, r) = 0$, the general partial differential equation involving the three independent variables x, y, z , and the derived functions p, q, r of the dependent variable u . The question is therefore to find an exact differential; for then writing $u = \int(pdx + qdy + rdz)$, we have the solution of the given partial differential equation.

Whatever be the method adopted, it comes out that the solution depends on the integration of the system of ordinary differential equations

$$\frac{dp}{dx} = \frac{dq}{dy} = \frac{dr}{dz} = \frac{dx}{dp} = \frac{dy}{dq} = \frac{dz}{dr},$$

and the answer consists first in showing this and, secondly, in showing how from an integral or integrals of the system we pass to the solution of the partial differential equation.

Considering the partial differential equation in the form actually proposed, and instead of ϕ write H , where H will stand for the given function of x, y, z, p, q, r , which is the arbitrary constant H ; making this change, and putting the foregoing equal to the arbitrary constant H ; making this change, and putting the foregoing equations equal to the differential dt of a new variable, the system of ordinary differential equations is

$$\begin{aligned} \frac{dp}{dt} &= -\frac{dH}{dx}, & \frac{dq}{dt} &= -\frac{dH}{dy}, & \frac{dr}{dt} &= -\frac{dH}{dz}, \\ \frac{dx}{dt} &= \frac{dH}{dp}, & \frac{dy}{dt} &= \frac{dH}{dq}, & \frac{dz}{dt} &= \frac{dH}{dr}, \end{aligned}$$

where H is a given function of x, y, z, p, q, r . This is, in fact, the Hamiltonian system of equations; and it was in view to the connexion that the partial differential equation was proposed in its actual form.

646.

ON THE GENERAL EQUATION OF DIFFERENCES OF THE SECOND ORDER.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 23–25.]

(Page 47.) CONSIDER the equation of differences

$$u_s = a_s u_{s-1} + b_{s-1} u_{s-2},$$

viz. we have

$$\begin{aligned} u_1 &= a_1 u_0 + b_0 u_{-1}, \\ u_2 &= a_2 u_1 + b_1 u_0, \\ u_3 &= a_3 u_2 + b_2 u_1, \\ u_4 &= a_4 u_3 + b_3 u_2, \\ u_5 &= a_5 u_4 + b_4 u_3, \\ &\&c., \end{aligned}$$

and thence

$$\begin{aligned} u_2 &= \begin{vmatrix} a_2 a_1 \\ +b_1 \end{vmatrix} u_1 + a_2 b_0 u_0, \\ u_3 &= \begin{vmatrix} a_3 a_2 a_1 \\ +a_3 b_1 \\ +a_1 b_2 \end{vmatrix} u_1 + \begin{vmatrix} a_3 + a_3 a_2 \\ +b_2 \end{vmatrix} b_0 u_0, \\ u_4 &= \begin{vmatrix} a_4 a_3 a_2 a_1 \\ +a_4 a_3 b_1 \\ +a_4 a_1 b_2 \\ +a_2 a_1 b_3 \\ +b_1 b_3 \end{vmatrix} u_1 + \begin{vmatrix} a_4 + a_4 a_3 a_2 \\ +a_4 b_2 \\ +a_2 b_3 \end{vmatrix} b_0 u_0, \end{aligned}$$

(Page 48.)

$$u_5 = \begin{vmatrix} a_5 a_4 a_3 a_2 a_1 \\ +a_5 a_4 a_3 b_1 \\ +a_5 a_4 a_1 b_2 \\ +a_5 a_2 a_1 b_3 \\ +a_3 a_2 a_1 b_4 \\ +a_5 b_1 b_3 \\ +a_3 b_1 b_4 \\ +a_1 b_2 b_4 \end{vmatrix} u_1 + \begin{vmatrix} a_5 a_4 a_3 a_2 \\ +a_5 a_4 b_2 \\ +a_5 a_2 b_3 \\ +a_3 a_2 b_4 \\ +b_2 b_4 \end{vmatrix} b_0 u_0, \\ \&c.$$

It is now easy to see the law; viz. writing for instance

$$u_5 = 54321 \cdot u_1 + 5432 \cdot b_0 u_0,$$

then 54321 has a leading term $a_5 a_4 a_3 a_2 a_1$: it has terms derived from this by changing any pair $a_5 a_4, a_4 a_3$, into b_4, b_3, b_2, b_1 etc.; it has terms derived by changing two pairs $a_5 a_4, a_3 a_2$ into $b_4 b_3, b_2 b_1$; and so on; where observe that the expression of two consecutive a 's.

And, similarly, 5432 has a leading term $a_5 a_4 a_3 a_2$; the other terms being derived from this in the same manner precisely.

The solution of $u_s = la(au_{s-1} - u_{s-2})$ is included in, and might be deduced from the foregoing, but it is convenient to obtain it separately. Supposing for greater simplicity that $u_{-1} = 0, u_0 = 1$ (or, what is the same thing, $u_0 = 1, u_1 = l_1 a$), we find

$$\begin{aligned} u_0 &= 1, \\ u_1 &= l_1 a, \\ u_2 &= l_2 l_1 a^2 - l_2, \\ u_3 &= l_3 l_2 l_1 a^3 - \begin{vmatrix} l_3 l_2 \\ +l_3 l_1 \end{vmatrix} a, \\ u_4 &= l_4 l_3 l_2 l_1 a^4 - \begin{vmatrix} l_4 l_3 l_2 \\ +l_4 l_3 l_1 \\ +l_4 l_2 l_1 \end{vmatrix} a^2 + l_4 l_2, \\ u_5 &= l_5 l_4 l_3 l_2 l_1 a^5 - \begin{vmatrix} l_5 l_4 l_3 l_2 \\ +l_5 l_4 l_3 l_1 \\ +l_5 l_4 l_2 l_1 \\ +l_5 l_3 l_2 l_1 \end{vmatrix} a^3 + \begin{vmatrix} l_5 l_3 l_2 \\ +l_5 l_2 l_1 \end{vmatrix} a, \end{aligned}$$

&c.,

(Page 49.) viz. we may for example write

$$u_5 = l_5 \cdot 4321 \cdot a^5 - 4321 \cdot () a^3 + 4321 \cdot () a,$$

where

$$4321 \text{ denotes } l_4 l_3 l_2 l_1;$$

in 4321() we omit successively each number, viz. we thus obtain

$$432 + 431 + 421 + 321, \quad = l_4 l_3 l_2 + l_4 l_3 l_1 + l_4 l_2 l_1 + l_3 l_2 l_1;$$

in 4321(·) we omit successively each pair of two non-consecutive numbers, viz. the omitted numbers being 1, 3; 1, 4; 2, 4, we obtain

$$42 + 32 + 31, \quad = l_4 l_2 + l_3 l_2 + l_3 l_1;$$

and so on, the omissions being each four numbers, &c. two of them consecutive; thus in 654321(·) the omissions are 5, 3, 1; 4, 6, 2, or two of them consecutive; or the symbol is

$$642 + 531 + 521 + 421;$$

and so on, the omissions being each four numbers, &c. two of them consecutive;

$$642 + 531 + 21,$$

$$= l_6 l_4 l_2 + l_5 l_3 l_1.$$

As an application, a solution of the differential equation $\frac{d}{dx} \left(x \frac{dy}{dx} \right) + (x - a)y = 0$ is $y = u_0 + u_1 x + u_2 x^2 + \&c.$, where $s^2 u_s = a u_{s-1} - u_{s-2}$, and in particular $1^2 u_1 = a u_0$, the equation of differences is thus of the form in question, and retaining l_s in place of its value, $= s^2$, the solution is $u_0 = 1$, $u_1 = l_1 a$, $u_2 = l_2 l_1 a^2 - l_2$, &c. *ut supra*. The differential equation was considered by the Rev. H. J. Sharpe, who mentioned it to Prof. Stokes.

647.

ON THE QUARTIC SURFACES REPRESENTED BY THE EQUATION, SYMMETRICAL DETERMINANT = 0.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 46–52.]

(Page 50.) CONSIDER the equation

$$\nabla = \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix} = 0,$$

where for the moment $(a, b, \dots)$ denote linear functions of the coordinates (x, y, z, w) . This is a quartic surface having 10 nodes; viz. if we write $(A, B, \dots)$ for the first minors of the determinant, then the cubic surfaces $A = 0$, $B = 0$, $\dots$ have in common 10 points which are nodes of the quartic surface.

Suppose that (a, b, c, f, g, h) are linear functions of the form (x, y) ; then, observing that every term of ∇ contains as a factor

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

or one of its first minors, it is clear that the line $x = 0$, $y = 0$ is a double line on the surface. But the number of nodes is now less than 10; in fact, writing $(x = 0, y = 0)$, we make each of the first minors $A = 0$, $B = 0$, $\dots$ have in common 10 points which are nodes of the quartic surface, which is common to the line $(x = 0, y = 0)$, and there is a diminution in the number of their common intersections. I do not pursue the enquiry, but pass to a different question.

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one of these is θ , so that $\theta^n = a$, then, assuming $x = \theta y$, we have $y^n - 1 = 0$, which equation (like the more general one $x^n - a = 0$) has precisely n roots; it is shown that these are $1, \omega, \omega^2, \dots, \omega^{n-1}$, where ω is a complex number $\alpha + \beta i$ such that $\alpha^2 + \beta^2 = 1$, or, what is the same thing, a complex number of the form $\cos \theta + i \sin \theta$; and it then at once appears that θ may be taken $= \frac{2\pi}{n}$. We have thus the trigonometrical solution of the equation $x^n - 1 = 0$. We may also obtain a like trigonometrical solution of the first-mentioned equation $x^n - a = 0$. We are thus led to the notion (a numerical) of the radical $a^{\frac{1}{n}}$, regarded as an n -valued function, viz. any one of these being denoted by $\sqrt[n]{a}$, then the series of values is

$$\sqrt[n]{a}, \omega \sqrt[n]{a}, \dots, \omega^{n-1} \sqrt[n]{a}.$$

Or we may, if we please, use $\sqrt[n]{a}$, instead of $a^{\frac{1}{n}}$, as a symbol to denote the n -valued function.

It is not necessary, as regards the equation $x^n - 1 = 0$, to refer here to the distinctions between the cases n a prime, and a composite, number.

As the coefficients of an algebraical equation may be numerical, all which follows in regard to algebraical equations, is (with, it may be, some few modifications) applicable to numerical equations; and hence, concluding for the present this subject, it will be convenient to pass on to algebraical equations.

II. We consider, secondly, an algebraical equation

$$x^n - p_1 x^{n-1} + \dots = 0,$$

and we here *assume* the existence of roots, viz. we assume that there are n quantities $a, b, c, \dots$ (in general, all of them different, but in particular cases they may become equal in sets in any manner), such that

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots$$

Or, looking at the question in a different point of view, and starting with the roots $a, b, c, \dots$ as given, we express the product of the n factors $x - a, x - b, \dots$ in the foregoing form, and thus arrive at an equation of the order n having the n roots $a, b, c, \dots$. In either case, we have

$$p_1 = \Sigma a, p_2 = \Sigma ab, \dots, p_n = abc \dots,$$

viz. regarding the coefficients $p_1, p_2, \dots, p_n$ as given, then we assume the existence of roots $a, b, c, \dots$ such that $p_1 = \Sigma a$, &c., or regarding the roots as given, then we write p_1, p_2 , &c., to denote the functions $\Sigma a, \Sigma ab$, &c.

It is to be noticed that, in virtue of

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b), \text{ \&c.,}$$

or of the equivalent equations $p_1 = \Sigma a$, &c., then

$$\begin{aligned} a^n - p_1 a^{n-1} + \dots &= 0, \\ b^n - p_1 b^{n-1} + \dots &= 0, \\ &\&c., \end{aligned}$$

(viz. it is for this reason that $a, b, \dots$ are said to be roots of $x^n - p_1 x^{n-1} + \dots = 0$); and, moreover, that conversely from the last-mentioned equations, assuming that $a, b, \dots$ are all different, we deduce

$$p_1 = \Sigma a, \quad p_2 = \Sigma ab, \quad \&c.,$$

and

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots$$

Observe that, if for instance $a = b$, then the two equations $a^n - p_1 a^{n-1} + \dots = 0$, $b^n - p_1 b^{n-1} + \dots = 0$ would reduce themselves to a single equation, which would not of itself express that a was a double root, that is, that $(x - a)^2$ was a factor of $x^n - p_1 x^{n-1} + \dots$; but by considering b as the limit of $a + h$, h indefinitely small, we obtain a second equation

$$na^{n-1} - (n-1)p_1 a^{n-2} + \dots = 0,$$

which, with the first, expresses that a is a double root; and then the whole system of equations leads, as before, to the equations $p_1 = \Sigma a$, &c. But this in passing: the general case is when the roots are all unequal.

We have then the *theorem* that every rational symmetrical function of the roots is a rational function of the coefficients; this is an easy consequence from the less general theorem, every rational and integral symmetrical function of the roots is a rational and integral function of the coefficients.

In particular, the sums of powers Σa^2 , Σa^3 , &c., are rational and integral functions of the coefficients.

An ordinary process, as regards the expression of other functions $\Sigma a^2 b^2$, &c., in terms of the coefficients, is to make them depend on the functions Σa^2 , &c., but this is *very objectionable*; the true theory consists in showing that we have systems of equations

$$\begin{aligned} p_1 &= \Sigma a, \\ \left\{ \begin{aligned} p_2 &= \Sigma ab, \\ p_1^2 &= \Sigma a^2 + 2\Sigma ab, \end{aligned} \right. \\ \left\{ \begin{aligned} p_3 &= \Sigma abc, \\ p_1 p_2 &= \Sigma a^2 b + 3\Sigma abc, \\ p_1^3 &= \Sigma a^3 + 3\Sigma a^2 b + 6\Sigma abc, \end{aligned} \right. \\ &\&c., \quad \&c. \end{aligned}$$

where, in each system, there are precisely as many equations as there are root-functions on the right-hand side, e.g. 3 equations and 3 functions Σabc , $\Sigma a^2 b$, Σa^3 . Hence, in each system, the root-functions can be determined linearly in terms of the powers and products of the coefficients.

It follows that it is possible to determine an equation (of an assignable order) having for roots any given (unsymmetrical) functions of the roots of a given equation. For example, in the case of a quartic equation, roots (a, b, c, d) , it is possible to find an equation having the roots ab, ac, ad, bc, bd, cd , being therefore a sextic equation; viz. in the product $(y - ab)(y - ac)(y - ad)(y - bc)(y - bd)(y - cd)$, the coefficients of the several powers of y will be symmetrical functions of a, b, c, d , and therefore rational and integral functions of the coefficients of the original quartic equation.

In connexion herewith, the question arises as to the number of values (obtained by permutations of the roots) of given unsymmetrical functions of the roots; for instance, with roots (a, b, c, d) as before, how many values are there of the function $ab + cd$; or, better, how many functions are there of this form; the answer is 3, viz. $ab + cd, ac + bd, ad + bc$; or, again, we may ask whether it is possible to obtain functions of a given number of values, 3-valued, 4-valued functions, &c.

We have, moreover, the very important *theorem* that, given the value of any unsymmetrical function, e.g. $ab + cd$, it is in general possible to determine rationally the value of any similar function, e.g. $(a + b)^3 + (c + d)^3$.

The *à priori* ground of this theorem may be illustrated by means of a numerical equation. Suppose, e.g. that the roots of a quartic equation are 1, 2, 3, 4; then if it is given that $ab + cd = 14$, this in effect determines a, b to be 1, 2 (viz. $a = 1, b = 2$, or else $a = 2, b = 1$) and c, d to be 3, 4 (viz. $c = 3, d = 4$, or else $c = 4, d = 3$); and it therefore in effect determines $(a + b)^3 + (c + d)^3$ to be $= 370$, and not any other value. And we can in the same way account for cases of failure as regards particular equations; thus, the roots being 1, 2, 3, 4, as above, $a^2b = 2$ determines a to be $= 1$ and b to be $= 2$; but if the roots had been 1, 2, 4, 16, then $a^2b = 16$ does not uniquely determine a and b , but only makes them to be 1 and 16, or else 2 and 4, respectively.

As to the *à posteriori* proof, assume, for instance, $t_1 = ab + cd, y_1 = (a + b)^3 + (c + d)^3$, and so $t_2 = ac + db, y_2 = (a + c)^3 + (d + b)^3$, &c.—in the present case there are only the functions t_1, t_2, t_3 and y_1, y_2, y_3 —then $y_1 + y_2 + y_3, t_1y_1 + t_2y_2 + t_3y_3, t_1^2y_1 + t_2^2y_2 + t_3^2y_3$ will be respectively symmetrical functions of the roots of the quartic, and therefore rational and integral functions of its coefficients, that is, they will be known.

Imagine, in the first instance, that t_1, t_2, t_3 are all known; then the equations being linear in y_1, y_2, y_3 , these can be expressed rationally in terms of known functions of the coefficients and of t_1, t_2, t_3 , that is, y_1, y_2, y_3 will be known. But observe further, that y_1 is obtained as a function of t_1, t_2, t_3 symmetrical as regards t_2, t_3 ; it can consequently be expressed as a rational function of t_1 and of $t_2 + t_3, t_2t_3$, or, what is the same thing, of t_1 and $t_1 + t_2 + t_3, t_1t_2 + t_1t_3 + t_2t_3, t_1t_2t_3$; but these last will be symmetrical functions of the roots, and as such expressible rationally in terms of the coefficients: that is, y_1 will be expressed as a rational function of t_1 and of the coefficients, or, t_1 being known, y_1 will be rationally determined.

We may consider now the question of the algebraical solution of equations, or, more accurately, that of the *solution of equations by radicals*.

In the case of a quadric equation $x^2 + px + q = 0$, we can find for x , by the assistance of the sign $\sqrt{(\)}$ or $(\)^{\frac{1}{2}}$, an expression for x as a two-valued function of the coefficients p, q , such that, substituting this value in the equation, the equation is thereby identically satisfied, viz. we have

$$x = -\frac{1}{2}p \pm \sqrt{\left(\frac{1}{4}p^2 - q\right)},$$

giving

$$\begin{array}{rcl} x^2 & = & \frac{1}{4}p^2 - q \mp p\sqrt{\left(\frac{1}{4}p^2 - q\right)} \\ + px & = & -\frac{1}{2}p^2 \pm p\sqrt{\left(\frac{1}{4}p^2 - q\right)} \\ + q & = & + q \\ \hline x^2 + px + q & = & 0, \end{array}$$

and the equation is on this account said to be algebraically solvable, or, more accurately, to be *solvable by radicals*. Or we may, by writing $x = -\frac{1}{2}p + z$, reduce the equation to $z^2 = \frac{1}{4}p^2 - q$, viz. to an equation of the form $z^2 = a$, and, in virtue of its being thus reducible, we may say that the equation is solvable by radicals. And the question for an equation of any higher order is, say of the order n , can we by means of radicals,

that is, by aid of the sign $\sqrt[m]{(\)}$ or $(\)^{\frac{1}{m}}$, using as many as we please of such signs and with any values of m , find an n -valued function (or any function) of the coefficients, which substituted for x in the equation shall satisfy it identically.

It will be observed that the coefficients $p, q, \dots$ are not explicitly considered as numbers, but that even if they do denote numbers, the question whether a numerical equation admits of solution by radicals is wholly unconnected with the before-mentioned theorem of the existence of the n roots of such an equation. It does not even follow that, in the case of a numerical equation solvable by radicals, the algebraical expression of x gives the numerical solution; but this requires explanation. Consider, first, a numerical quadric equation with imaginary coefficients; in the formula $x = -\frac{1}{2}p \pm \sqrt{\left(\frac{1}{4}p^2 - q\right)}$, substituting for p, q their given numerical values we obtain for x an expression of the form $x = \alpha + \beta i \pm \sqrt{(\gamma + \delta i)}$, where $\alpha, \beta, \gamma, \delta$ are real numbers; this value substituted in the numerical equation would satisfy it identically and it is thus an algebraical solution; but there is no obvious *a priori* reason why the expression $\sqrt{(\gamma + \delta i)}$ should have a value $= c + di$, where c and d are real numbers calculable by the extraction of a root or roots of real numbers; it appears upon investigation that $\sqrt{(\gamma + \delta i)}$ has such a value calculable by means of the radical expression $\sqrt{\{\sqrt{(\gamma^2 + \delta^2)} \pm \gamma\}}$; and hence that the algebraical solution of a quadric equation does in every case give the numerical solution of a numerical quadric. The case of a numerical cubic will be considered presently.

A cubic equation can be solved by radicals, viz. taking for greater simplicity the cubic in the reduced form $x^3 - qx - r = 0$, and writing $x = a + b$, this will be a solution if only $3ab = q$, and $a^3 + b^3 = r$, or say $\frac{1}{2}(a^3 + b^3) = \frac{1}{2}r$; whence

$$\frac{1}{2}(a^3 - b^3) = \pm \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)},$$

and therefore

$$a = \sqrt[3]{\frac{1}{2}r \pm \sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)}},$$

a six-valued function of q, r . But then writing $b = \frac{q}{3a}$, we have, as may be shown, $a + b$ a three-valued function of the coefficients; it would have been wrong to complete the solution by writing $b = \sqrt[3]{\frac{1}{2}r \pm \sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)}}$, since here $(a + b)$ would be given as a 9-valued function, having only 3 of its values roots, and the other 6 values being irrelevant. An interesting variation of the solution is to write $x = ab(a + b)$, giving $a^3b^3(a^3 + b^3) = r$ and $3a^3b^3 = q$, or say $\frac{1}{2}(a^3 + b^3) = \frac{3}{2}\frac{r}{q}$, $a^3b^3 = \frac{1}{3}q$; whence

$$\{\frac{1}{2}(a^3 - b^3)\}^2 = \frac{9}{q^2}(\frac{1}{4}r^2 - \frac{1}{27}q^3),$$

and therefore

$$a = \sqrt[3]{\left\{\frac{3}{2}\frac{r}{q} \pm \frac{3}{q}\sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)}\right\}}, \quad b = \sqrt[3]{\left\{\frac{3}{2}\frac{r}{q} \mp \frac{3}{q}\sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)}\right\}},$$

and here although a, b are each of them a 6-valued function, yet, as may be shown, $ab(a + b)$ is only a 3-valued function.

In the case of a numerical cubic, even when the coefficients are real, substituting their values in the expression

$$x = \sqrt[3]{\frac{1}{2}r \pm \sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)}} + [\frac{1}{3}q \div \sqrt[3]{\frac{1}{2}r \pm \sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)}}],$$

this *may* depend on an expression of the form $\sqrt[3]{(\gamma + \delta i)}$, where γ and δ are real numbers (viz. it will do so if $\frac{1}{4}r^2 - \frac{1}{27}q^3$ is a negative number), and here we *cannot* by the extraction of any root or roots of real numbers reduce $\sqrt[3]{(\gamma + \delta i)}$ to the form $c + di$, c and d real numbers; hence, here the algebraical solution does not give the numerical solution. It is to be added that the case in question, called the "irreducible case," is that wherein the three roots of the cubic equation are all real; if the roots are one real and two imaginary, then, contrariwise, the quantity under the cube root is real, and the algebraical solution gives the numerical one.

The irreducible case is solvable by a trigonometrical formula, but this is not a solution by radicals; it consists, in effect, in reducing the given numerical cubic (not to a cubic of the form $x^3 = a$, solvable by the extraction of a cube root, but) to a cubic of the form $4x^3 - 3x = a$, corresponding to the equation $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ which serves to determine $\cos\theta$ when $\cos 3\theta$ is known.

A quartic equation is solvable by radicals; and it may be remarked, that the existence of such a solution depends on the existence of 3-valued functions such as $ab + cd$, of the four roots (a, b, c, d) ; by what precedes, $ab + cd$ is the root of a cubic equation, which equation is solvable by radicals; hence $ab + cd$ can be found by radicals; and since $abcd$ is a given value, ab and cd can each be found by radicals. But by what precedes, if ab be known, then any similar function, say $a + b$, is obtainable rationally; and, consequently, from the values of $a + b$ and ab we may by radicals

obtain the value of a or b , that is, an expression for a root of the given quartic expression; the expression finally obtained is 4-valued, corresponding to the different values of the several radicals which enter therein, and we have therefore the expression by radicals of each of the four roots of the quartic equation. But when the quartic is numerical, the same thing arises as in the cubic: the algebraical expression does not in every case give the numerical one.

It will be understood from the foregoing explanation as to the quartic, how in the next following case, that of a quintic equation, the question of the solvability by radicals depends on the existence or non-existence of k -valued functions of the five roots (a, b, c, d, e); a fundamental theorem on the subject is that a rational function of 5 letters, if it has less than 5, cannot have more than 2 values; viz. that there are no 3-valued, or 4-valued, functions of 5 letters; and by reasoning, depending in part upon this theorem, Abel showed that a general quintic equation is not solvable by radicals: and *a fortiori* the general equation of any order higher than 5 is not solvable by radicals.

The general theory of the solvability of an equation by radicals depends very much on Vandermonde's remark, that supposing an equation is solvable (by radicals) and that we have therefore an algebraical expression of x in terms of the coefficients, then substituting for the coefficients their values in terms of the roots, the resulting value of the expression must reduce itself to any one at pleasure of the roots $a, b, c, \dots$; thus in the case of the quadric equation where the solution is $x = +\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)}$, writing for p, q their values $a+b, ab$, this is $x = \frac{1}{2}[(a+b) \pm \sqrt{(a-b)^2}]$, $= a$ or b according to the value of the radical. But it is not considered necessary in the present sketch to go further into the theory of the solvability of an equation by radicals. It may be proper to remark that, for quintic equations, there are solutions analogous to the trigonometrical solution of a cubic equation, viz. the quintic equation is here in effect reduced to some special form of quintic equation; for instance, to Jerrard's form $x^5 + ax + b = 0$ or to some form presenting itself in the theory of elliptic functions; but the solutions in question are not solutions by radicals. And there are various other interesting parts of the theory which have been excluded from consideration.

632.

ON ARONHOLD'S INTEGRATION-FORMULA.

[From the *Messenger of Mathematics*, vol. v. (1876), pp. 88—90.]

THE fundamental theorem in Aronhold's Memoir, "Ueber eine neue algebraische Behandlungsweise der Integrale... $\Pi(x, y)dx$, &c.," *Crelle*, t. LXI. (1863), pp. 95—145, is a theorem of *indefinite* integration. The form is

$$\Delta \int \frac{dx}{(\alpha x + \beta y + \gamma)(hx + by + f)} = \log \frac{(a\xi + h\eta + g)x + (h\xi + b\eta + f)y + g\xi + f\eta + c}{\alpha x + \beta y + \gamma},$$

where y is a certain irrational function of x , determined by a quadric equation, and the other symbols denote constants connected by certain relations; viz. writing, for shortness,

$$U = (a, b, c, f, g, h\chi x, y, 1)^2, = (a, \dots \chi x, y, 1)^2 \text{ for shortness,}$$

that is,

$$= ax^2 + 2hxy + by^2 + 2fy + 2gx + c;$$

$$W = (a, b, c, f, g, h\chi x, y, 1\chi\xi, \eta, 1), = (a, \dots \chi x, y, 1\chi\xi, \eta, 1),$$

that is,

$$= (ax + hy + g)\xi + (hx + by + f)\eta + gx + fy + c,$$

or

$$(a\xi + h\eta + g)x + (h\xi + b\eta + f)y + g\xi + f\eta + c;$$

$$(P, Q, R) = (ax + hy + g, hx + by + f, gx + fy + c),$$

$$(P_0, Q_0, R_0) = (a\xi + h\eta + g, h\xi + b\eta + f, g\xi + f\eta + c),$$

$$\Omega = \alpha x + \beta y + \gamma,$$

$$\Omega_0 = \alpha\xi + \beta\eta + \gamma,$$

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

then y is determined as a function of x by the equation $U=0$, that is,

$$(a, b, c, f, g, h \chi x, y, 1)^2 = 0;$$

or, what is the same thing,

$$by = -\{hx + f + \sqrt{(-Cx^2 + 2Gx - A)}\};$$

the constants α, β, ξ, η are such that

$$(a, b, c, f, g, h \chi \xi, \eta, 1)^2 = 0,$$

$$\alpha \xi + \beta \eta + \gamma = 0,$$

that is,

$$\Omega_0 = 0;$$

and the value of Λ is given by

$$\Lambda^2 = -(A, B, C, F, G, H \chi \alpha, \beta, \gamma)^2.$$

The theorem may therefore be written

$$\Lambda \int \frac{dx}{\Omega Q} = \log \frac{W}{\Omega},$$

where the several symbols have the significations explained above.

The verification is as follows. We ought to have

$$\frac{\Lambda dx}{\Omega Q} = \frac{P_0 dx + Q_0 dy}{W} - \frac{\alpha dx + \beta dy}{\Omega},$$

when dx, dy satisfy the relation $P dx + Q dy = 0$, viz. substituting for dy the value $-\frac{P dx}{Q}$, the equation becomes

$$\frac{\Lambda}{\Omega} = \frac{P_0 Q - P Q_0}{W} - \frac{\alpha Q - \beta P}{\Omega},$$

that is, substituting for Ω its value,

$$\Lambda W = (P_0 Q - P Q_0)(\alpha x + \beta y + \gamma) - (\alpha Q - \beta P) W.$$

On the right-hand side, substituting for W its value,

$$\text{coeff. } \alpha = x(P_0 Q - P Q_0) - Q(P_0 x + Q_0 y + R_0), \quad = Q_0 R - Q R_0,$$

$$\text{coeff. } \beta = y(P_0 Q - P Q_0) + P(P_0 x + Q_0 y + R_0), \quad = R_0 P - R P_0,$$

(as at once appears by aid of the relation $U = P x + Q y + R = 0$),

$$\text{coeff. } \gamma = P_0 Q - P Q_0.$$

The equation to be verified thus is

$$\Lambda W = \begin{vmatrix} \alpha & \beta & \gamma \\ P_0 & Q_0 & R_0 \\ P & Q & R \end{vmatrix},$$

which, substituting therein for P, Q, R, P_0, Q_0, R_0 , their values, and writing

$$(\lambda, \mu, \nu) = (\eta - y, x - \xi, \xi y - \eta x),$$

is in fact

$$\Lambda W = (A, \dots \check{\lambda}, \mu, \nu \check{\alpha}, \beta, \gamma).$$

We have identically

$$(a, \dots \check{x}, y, 1)^2 \cdot (a, \dots \check{\xi}, \eta, 1)^2 - W^2 = (A, \dots \check{\lambda}, \mu, \nu)^2,$$

which, in virtue of $(a, \dots \check{\xi}, \eta, 1)^2 = 0$, gives

$$W^2 = -(A, \dots \check{\lambda}, \mu, \nu)^2;$$

and since $\Lambda^2 = -(A, \dots \check{\alpha}, \beta, \gamma)^2$, the equation is thus

$$\sqrt{-(A, \dots \check{\alpha}, \beta, \gamma)^2} \cdot \sqrt{-(A, \dots \check{\lambda}, \mu, \nu)^2} = (A, \dots \check{\lambda}, \mu, \nu \check{\alpha}, \beta, \gamma),$$

that is,

$$(A, \dots \check{\alpha}, \beta, \gamma)^2 \cdot (A, \dots \check{\lambda}, \mu, \nu)^2 - [(A, \dots \check{\lambda}, \mu, \nu \check{\alpha}, \beta, \gamma)]^2 = 0.$$

The left-hand side is here identically

$$= K(a, \dots \check{\gamma}\mu - \beta\nu, \alpha\nu - \gamma\lambda, \beta\lambda - \alpha\mu)^2:$$

substituting for λ, μ, ν their values, we find

$$(\gamma\mu - \beta\nu, \alpha\nu - \gamma\lambda, \beta\lambda - \alpha\mu) = (x\Omega_0 - \xi\Omega, y\Omega_0 - \eta\Omega, z\Omega_0 - \zeta\Omega);$$

viz. in virtue of $\Omega_0 = 0$, these are $= -\xi\Omega, -\eta\Omega, -\zeta\Omega$, and the quadric function is $= K\Omega^2(a, \dots \check{\xi}, \eta, 1)^2$, vanishing in virtue of the relation $(a, \dots \check{\xi}, \eta, 1)^2 = 0$.

The equation in question

$$\sqrt{-(A, \dots \check{\alpha}, \beta, \gamma)^2} \cdot \sqrt{-(A, \dots \check{\lambda}, \mu, \nu)^2} = (A, \dots \check{\lambda}, \mu, \nu \check{\alpha}, \beta, \gamma)$$

is thus verified, and the theorem is proved.

633.

NOTE ON MR MARTIN'S PAPER, "ON THE INTEGRALS OF
SOME DIFFERENTIALS."

[From the *Messenger of Mathematics*, vol. v. (1876), p. 163.]

The Note refers to a detail in a process of integration.

634.

THEOREMS IN TRIGONOMETRY AND ON PARTITIONS.

[From the *Messenger of Mathematics*, vol. v. (1876), p. 164, and p. 188.]

If

$$A + B + C + F + G + H = 0,$$

then

$$\begin{vmatrix} \sin \overline{A + F} \sin \overline{B + F} \sin \overline{C + F}, & \cos F, & \sin F \\ \sin \overline{A + G} \sin \overline{B + G} \sin \overline{C + G}, & \cos G, & \sin G \\ \sin \overline{A + H} \sin \overline{B + H} \sin \overline{C + H}, & \cos H, & \sin H \end{vmatrix} = 0.$$

Let u_n = number of partitions of n , no part less than 2, the order attended to; e.g. if $n=7$, the partitions are 7, 52, 25, 43, 34, 322, 232, 223, $u_7=8$; the series is

$$u_2 = 1,$$

$$u_3 = 1,$$

$$u_4 = 2,$$

$$u_5 = 3,$$

$$u_6 = 5,$$

$$u_7 = 8,$$

$$u_8 = 13,$$

$$u_9 = 21,$$

where each term is the sum of the next preceding two terms.

635.

NOTE ON THE DEMONSTRATION OF CLAIRAUT'S THEOREM.

[From the *Messenger of Mathematics*, vol. v. (1876), pp. 166, 167.]

It seems worth while to indicate what the leading steps of the demonstration are.

The potential of the Earth's mass upon an external or superficial point is taken to be

$$V, = \frac{V_0}{r} + \frac{V_1}{r^2} + \frac{V_2}{r^3} + \&c.,$$

where $V_1, V_2, V_3, \dots$ are Laplace's functions of the angular coordinates.

The surface is assumed to be a nearly spherical surface $r = a(1 + u)$, where $u = u_1 + u_2 + \&c.$, and $u_1, u_2, \dots$ are Laplace's functions of the angular coordinates. To be a surface of equilibrium, with an equation $V + \frac{1}{2}\omega^2 r^2 \sin^2 \theta = C$, the latter must be equivalent to the equation $r = a(1 + u)$, and it follows that we have

$$\begin{aligned} V_1 &= V_0 a u_1, \\ V_2 &= V_0 a^2 u_2 - \frac{1}{2} \omega^2 a^3 \left(\frac{1}{3} - \cos^2 \theta \right), \\ V_3 &= V_0 a^3 u_3, \\ &\&c., \end{aligned}$$

which values are to be substituted in the expression for V .

The whole force of gravity (due to the attraction and the centrifugal force) is taken to be $g, = -\frac{d}{dr} (V + \frac{1}{2}\omega^2 r^2 \sin^2 \theta)$, and it follows that

$$g = \frac{V_0}{a^2} (1 + u_2 + 2u_3 + \dots) - \frac{2}{3} \omega^2 a - \frac{5}{2} \omega^2 a \left(\frac{1}{3} - \cos^2 \theta \right),$$

which is of the form

$$g = G \left\{ 1 + u_2 - \frac{5}{2} \frac{\omega^2 a}{G} \left(\frac{1}{3} - \cos^2 \theta \right) + 2u_3 + \dots \right\}.$$

Taking the Earth to be the spheroid of revolution

$$r = a \{ 1 + \epsilon \left(\frac{1}{3} - \cos^2 \theta \right) \},$$

then

$$u_2 = \epsilon \left(\frac{1}{3} - \cos^2 \theta \right), \quad u_3 = 0, \quad \&c.,$$

and the equation is

$$g = G \left\{ 1 - \left(\frac{5}{2} \frac{\omega^2 a}{G} - \epsilon \right) \left(\frac{1}{3} - \cos^2 \theta \right) \right\},$$

or say

$$g = G \{ 1 - \left(\frac{5}{2} m - \epsilon \right) \left(\frac{1}{3} - \cos^2 \theta \right) \},$$

where $m, = \frac{\omega^2 a}{G}$, is the ratio of the centrifugal force at the equator to the force of gravity, which is the theorem in question. The expression "it follows" has been twice used as meaning it follows as a mere analytical consequence, in the proper degree of approximation, the steps of the deduction being purposely omitted.

636.

ON THE THEORY OF THE SINGULAR SOLUTIONS OF DIFFERENTIAL EQUATIONS OF THE FIRST ORDER.

[From the *Messenger of Mathematics*, vol. VI. (1877), pp. 23—27.]

IN continuation of the former paper with this title (*Messenger*, vol. II, 1873, pp. 6—12, [545]), I propose to discuss various particular examples, chiefly of cases in which the differential equation is of the form $(L, M, N \chi p, 1)^2 = 0$, where L, M, N are rational and integral functions of (x, y) , and whether it admits or does not admit of an integral equation $(P, Q, R \chi c, 1)^2 = 0$, where P, Q, R are rational and integral functions of (x, y) .

The singular solution of the differential equation

$$(L, M, N \chi p, 1)^2 = 0,$$

if there be a singular solution, is $S = 0$, where S is either $= LN - M^2$, or a factor of $LN - M^2$. But in general $LN - M^2$ is an indecomposable function, such that $LN - M^2 = 0$ is not a solution of the differential equation, and this being so, there is no singular solution; viz. a differential equation $(L, M, N \chi p, 1)^2 = 0$, where L, M, N are rational and integral functions of (x, y) , has not in general any singular solution.

Consider now a system of algebraical curves $U = 0$, where U is as regards (x, y) a rational and integral function of the order m , and depends in any manner on an arbitrary parameter C^* . I say that there is always a proper envelope, which envelope is the singular solution of the differential equation obtained by the elimination of C from the equation $U = 0$, and the derived equation in regard to (x, y) . It follows that the differential equation $(L, M, N \chi p, 1)^2 = 0$, which has no singular solution, does not admit of an integral of the form in question $U = 0$, viz. an integral representing a system of algebraic curves.

* The expressions in the text may be understood as extending to the case where U is a function of any number (α) of constants $c_1, c_2, \dots, c_\alpha$, connected by an $(\alpha - 1)$ fold relation, U thus virtually depending on a single arbitrary parameter.

The theorem just referred to, that the system of algebraic curves $U=0$ has always an envelope, is an interesting theorem, which I proceed to prove. Assume that in general, that is, for an arbitrary value of the parameter, the equation $U=0$ represents a curve of the order m , with δ nodes and κ cusps (and therefore of the class n , with i inflexions and τ double tangents, the numbers $m, \delta, \kappa, n, \tau, i$ being connected by Plücker's equations); for particular values of the parameter, the values of δ and κ may be increased, or the curve may break up, but this is immaterial.

The consecutive curve $U + \delta c d_c U = 0$ is a curve of the same order m , with δ nodes and κ cusps, consecutive to the nodes and cusps of the original curve U , and the two curves intersect in m^2 points; but of these, there are 2 coinciding with each node, and 3 coinciding with each cusp of the curve $U=0$, as at once appears by drawing a curve with a node or a cusp, and the consecutive curve with a consecutive node or cusp; the number of the remaining intersections is $=m^2 - 2\delta - 3\kappa$, and the envelope is the locus of these $m^2 - 2\delta - 3\kappa$ points. Observe that the two curves have in common n^2 tangents; but of these, 2 coincide with each double tangent and 3 coincide with each stationary tangent of the curve $U=0$, viz. the number of the remaining common tangents is $=n^2 - 2\tau - 3i$ (which is $=m^2 - 2\delta - 3\kappa$); and that these $n^2 - 2\tau - 3i$ common tangents are indefinitely near to the $m^2 - 2\delta - 3\kappa$ common points respectively, and are in fact the tangents of the envelope at the $m^2 - 2\delta - 3\kappa$ points respectively. Now in an algebraic curve we have $m + n = m^2 - 2\delta - 3\kappa$, viz. the number $m^2 - 2\delta - 3\kappa$ cannot be $=0$, and we have therefore always an envelope the locus of the system of the $m^2 - 2\delta - 3\kappa$ points. It might be thought that the conclusion extends to transcendental curves; if this were so, the result would prove too much, viz. it would follow that a differential equation $(L, M, N \chi p, 1)^2 = 0$ without a singular solution had no general integral; but it will appear by an example that the theorem as to the envelope does not extend to transcendental curves.

Ex. 1.

$$p^2 - (1 - y^2) = 0, \text{ that is, } dy^2 - (1 - y^2) dx^2 = 0.$$

Here there is no algebraical integral, but there is a quasi-algebraical integral of the form $(P, Q, R \chi c, 1)^2 = 0$; viz. starting with the form $y = \sin(x + C)$ and expressing $\sin C$ and $\cos C$ rationally in terms of a new parameter, this is

$$c^2(y + \cos x) - 2c \sin x + (y - \cos x) = 0,$$

where the coefficients are one-valued functions of (x, y) . The discriminant of the differential equation in regard to p and that of the integral equation in regard to c are each $=y^2 - 1$, and we have a true singular solution $y^2 - 1 = 0$.

Ex. 2.

$$(1 - x^2)p^2 - (1 - y^2) = 0,$$

that is,

$$(1 - x^2)dy^2 - (1 - y^2)dx^2 = 0.$$

We have here an algebraic integral of the proper form, which is at once derived from the circular form

$$C = \cos^{-1}x + \cos^{-1}y$$

by changing the constant, viz. this is

$$c^2 - 2cxy - (1 - x^2 - y^2) = 0.$$

The two discriminants are here each $=(x^2 - 1)(y^2 - 1)$, and we have

$$(x^2 - 1)(y^2 - 1) = 0$$

as a true singular solution. The curves are in fact the system of conics (ellipses and hyperbolas) each touching the four lines $x = 1$, $x = -1$, $y = 1$, $y = -1$.

Ex. 3.

$$(1 - y^2)p^2 - 1 = 0, \text{ that is, } (1 - y^2)dy^2 - dx^2 = 0.$$

This is an extremely interesting example: the curve is the orthogonal trajectory of the system of sinusoids $y = \sin(x + c)$, which is the integral of Example 1; and we thus at once see that the real portion of the curve is wholly included between the lines $y = -1$, $y = +1$, being an infinite continuous curve, having a series of equidistant cusps alternately at the one and the other line, and obtained by the continued repetition of the finite portion included between two consecutive cusps on the same line. The discriminant of the differential equation equated to zero gives $y^2 - 1 = 0$, the equation of the two lines in question; but this does not satisfy the differential equation, and it is consequently not a singular solution; by what precedes, it appears that it is, in fact, a cusp-locus.

We thus see that the curves which represent the integral equation have no *real* envelope; but it is to be further shown that there is no imaginary envelope, and that the curve obtained by the elimination of the parameter is, in fact, made up of a (imaginary) node-locus and of the foregoing cusp-locus.

The curve is properly represented by taking x , y each of them a one-valued function of the parameter θ , viz. we may write

$$y = \cos \theta,$$

$$x = c + \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta.$$

In fact, these values give

$$\frac{dy}{d\theta} = -\sin \theta, \quad \frac{dx}{d\theta} = \frac{1}{2}(1 - \cos 2\theta) = \sin^2 \theta,$$

and therefore

$$p = -\frac{1}{\sin \theta} = \frac{-1}{\sqrt{1 - y^2}},$$

that is, $(1 - y^2)p^2 - 1 = 0$, the differential equation.

It is obvious that to a given value of the parameter there corresponds a single point of the curve; and it is to be shown that, conversely, to a given point of the curve corresponds in general a single value of the parameter.

Suppose the coordinates of the given point are $y = \cos \alpha$, $x = c + \frac{1}{2}\alpha - \frac{1}{2}\sin 2\alpha$, where α is a determinate quantity; then, to find θ , we have

$$\cos \theta = \cos \alpha, \quad 2\theta - \sin 2\theta = 2\alpha - \sin 2\alpha.$$

The first equation gives $\theta = 2m\pi \pm \alpha$, and the second equation then is

$$4m\pi \pm 2\alpha \mp \sin 2\alpha = 2\alpha - \sin 2\alpha;$$

viz. taking the upper signs, this is $4m\pi = 0$, giving $m = 0$ and $\theta = \alpha$; and, taking the lower signs, it is $m\pi = \alpha - \sin \alpha$, which, α being given, is not in general satisfied; hence to the given point there corresponds *only* the value α of the parameter θ . If, however, α is such that $\alpha - \sin \alpha$ is equal to a multiple of π , say $r\pi$, then the last-mentioned equation is satisfied by the value $m = r$, so that to the given point of the curve correspond the two values α and $2r\pi - \alpha$ of the parameter; these values are in general unequal, and the point is then a node; but they may be equal, viz. this is so if $\alpha = r\pi$ (the point on the curve being then $y = \cos r\pi = \pm 1$, $x = c + \frac{1}{2}r\pi$), and the point is then a cusp; showing what was known, that there are on each of the lines $y = -1$, $y = +1$, an infinite series of equidistant cusps.

More definitely, suppose $\alpha = r\pi \pm \beta$, where β is a root of the equation $2\beta - \sin 2\beta = 0$, then

$$\sin 2\alpha = \pm \sin 2\beta, \quad 2\alpha - \sin 2\alpha = 2r\pi \pm (2\beta - \sin 2\beta) = 2r\pi,$$

and to the given point on the curve correspond the two values α and $2r\pi - \alpha$ of the parameter. If $\beta = 0$, we have, as above, the cusps on the two lines $y = +1$, $y = -1$ respectively; but if β be an imaginary root of the equation $2\beta - \sin 2\beta = 0$, then we have an infinite series of nodes on the imaginary line $y = \cos r\pi \cos \beta$; and there are an infinite number of such lines corresponding to the different imaginary roots of the equation $2\beta - \sin 2\beta = 0$.

From the form in which the equation of the curve is given, we cannot directly form the equation of the envelope by equating to zero the discriminant in regard to the constant c ; but we may determine the intersections of the curve by the consecutive curve (corresponding to a value $c + \delta c$ of the constant), and thus determine the locus of these intersections.

Consider for a moment the curves belonging to the constants c, c_1 , and let θ, θ_1 be the values of the parameter θ belonging to the points of intersection; we have $\cos \theta = \cos \theta_1$, $4c + 2\theta - \sin 2\theta = 4c_1 + 2\theta_1 - \sin 2\theta_1$; we have $\theta_1 = 2r\pi + \theta$, but we cannot thereby satisfy the second equation; or else $\theta_1 = 2r\pi - \theta$, giving

$$4c + 2\theta - \sin 2\theta = 4c_1 + 4r\pi - 2\theta + \sin 2\theta,$$

that is, $2\theta - \sin 2\theta = 2c_1 - 2c + 2r\pi$; and we have thus corresponding to any given value of r a series of values of θ , viz. these are $\theta = r\pi + \beta$, where β is any root of the equation

$$2\beta - \sin 2\beta = 2c_1 - 2c.$$

In particular, taking $c_1 = c$, the intersections are given by $\theta = r\pi + \beta$, where β is any root of the equation $2\beta - \sin 2\beta = 0$; viz. we have thus an infinite number of intersections lying on each of the lines $y = \cos r\pi \cos \beta$. If $\beta = 0$, the intersections lie on the two lines $y = 1$, $y = -1$ respectively; if β be an imaginary root of the equation $2\beta - \sin 2\beta = 0$, then they lie on the imaginary lines $y = \cos r\pi \cos \beta$. But by what precedes, it is clear that in the former case the intersections are nothing else than the cusps on the lines $y = 1$, $y = -1$; and in the latter case nothing else than the nodes on the lines $y = \cos r\pi \cos \beta$; viz. there is no proper envelope, but instead thereof we have lines of cusps and of nodes.

Ex. 4.

$$(1 - y^2)p^2 - (1 - x^2) = 0,$$

that is,

$$(1 - y^2)dy^2 - (1 - x^2)dx^2 = 0.$$

I have not examined this; the curve is the series of orthogonal trajectories of the conics of Example 2, and the integral equation may be represented by $y = \cos \theta$, $x = \cos \phi$, where $c = (2\theta - \sin 2\theta) - (2\phi - \sin 2\phi)$.

Equating to zero the discriminant of the differential equation, we have $(1 - y^2)(1 - x^2) = 0$, viz. the four lines $x = 1$, $x = -1$, $y = 1$, $y = -1$; this is not an envelope, but a locus of cusps.

637.

ON A DIFFERENTIAL EQUATION IN THE THEORY OF ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VI. (1877), p. 29.]

IN the differential equation

$$Q^2 - Q \left(k + \frac{1}{k} \right) - 3 = 3(1 - k^2) \frac{dQ}{dk},$$

considered *Messenger*, t. IV., pp. 69 and 110, [594] and [597], writing $Q = x$ and $k + \frac{1}{k} = y$, the equation becomes

$$dy = \frac{3(y^2 - 4) dx}{3 + xy - x^2},$$

and we have, as a particular solution,

$$y = \frac{1}{4} \left(x^3 - 6x - \frac{3}{x} \right).$$

To verify this, observe that from the value of y

$$dy = \frac{3}{4x^2} (x^2 - 1)^2 dx, \quad 3 + xy - x^2 = \frac{1}{4} (x^3 - 1)(x^2 - 9),$$

and the equation becomes

$$\frac{3}{4x^2} (x^2 - 1)^2 = \frac{\frac{3}{16x^2} \{(x^4 - 6x^2 - 3)^2 - 64x^2\}}{\frac{1}{4} (x^3 - 1)(x^2 - 9)},$$

viz. this is

$$(x^2 - 1)^2 (x^2 - 9) = (x^4 - 6x^2 - 3)^2 - 64x^2,$$

which is right.

638.

ON A q -FORMULA LEADING TO AN EXPRESSION FOR E_1 .

[From the *Messenger of Mathematics*, vol. VI. (1877), pp. 63—66.]

It is to be shown that we have identically

$$(1 + 2q + 2q^4 + 2q^9 + \dots)^4 - 16 \left(\frac{q}{1 - q^2} + \frac{2q^2}{1 - q^4} + \frac{3q^3}{1 - q^6} + \dots \right) \\ = \frac{1 - 9q - 25q^3 + 49q^6 + 81q^{10} - \dots}{1 - q - q^3 + q^6 + q^{10} - \dots} \dots\dots\dots (A);$$

or, what is the same thing,

$$(1 - 2q + 2q^4 - 2q^9 + \dots)^4 - 16 \left(\frac{-q}{1 - q^2} + \frac{2q^2}{1 - q^4} - \frac{3q^3}{1 - q^6} + \dots \right) \\ = \frac{1 + 9q + 25q^3 + 49q^6 + 81q^{10} + \dots}{1 + q + q^3 + q^6 + q^{10} + \dots} \dots\dots\dots (B),$$

where the form (A) is that intended to be made use of, but the form (B) is rather more convenient for the demonstration.

We have

$$(1 - 2q + 2q^4 - 2q^9 + \dots)^4 = 1 + 8 \left\{ \frac{-q}{1 + q} + \frac{2q^2}{1 + q^2} - \frac{3q^3}{1 + q^3} + \dots \right\},$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. I., p. 239), taking the formula as there written down, and changing q into $-q$.

Also, if for a moment

$$X = 1 + q + q^3 + q^6 + q^{10} + \&c.,$$

and

$$X' = \frac{dX}{dq},$$

C. X.

so that

$$qX' = q + 3q^3 + 6q^5 + 10q^{10} + \&c.,$$

then

$$X + 8qX' = 1 + 9q + 25q^3 + 49q^5 + 81q^{10} + \&c.,$$

so that the right-hand side of (B) is

$$\frac{X + 8qX'}{X}, = 1 + 8q \frac{X'}{X}.$$

But (*Fund. Nova*, p. 185, *Ges. Werke*, t. I., p. 237),

$$X = \frac{1 - q^2}{1 - q} \cdot \frac{1 - q^4}{1 - q^3} \cdot \frac{1 - q^6}{1 - q^5} \dots,$$

so that

$$\begin{aligned} \frac{X'}{X} &= \frac{-2q}{1 - q^2} - \frac{4q^3}{1 - q^4} - \frac{6q^5}{1 - q^6} - \dots \\ &\quad + \frac{1}{1 - q} + \frac{3q^2}{1 - q^3} + \frac{5q^4}{1 - q^5} + \dots \end{aligned}$$

And the equation (B) intended to be proved thus becomes

$$\begin{aligned} &1 + 8 \left\{ \frac{-q}{1 + q} + \frac{2q^2}{1 + q^2} - \frac{3q^3}{1 + q^3} + \dots \right\} \\ &- 16 \left\{ \frac{-q}{1 - q^2} + \frac{2q^2}{1 - q^4} - \frac{3q^3}{1 - q^6} + \dots \right\} \\ &= 1 + 8q \left\{ \frac{-2q}{1 - q^2} - \frac{4q^3}{1 - q^4} - \frac{6q^5}{1 - q^6} - \dots \right. \\ &\quad \left. + \frac{1}{1 - q} + \frac{3q^2}{1 - q^3} + \frac{5q^4}{1 - q^5} + \dots \right\}; \end{aligned}$$

viz. omitting the terms unity, dividing by $8q$, and transposing, this is

$$\begin{aligned} &-\frac{1}{1 + q} + \frac{2q}{1 + q^2} - \frac{3q^2}{1 + q^3} + \dots \\ &+ \frac{2}{1 - q^2} - \frac{4q}{1 - q^4} + \frac{6q^2}{1 - q^6} - \dots \\ &+ \frac{2q}{1 - q^2} + \frac{4q^3}{1 - q^4} + \frac{6q^5}{1 - q^6} + \dots \\ &-\frac{1}{1 - q} - \frac{3q^2}{1 - q^3} - \frac{5q^4}{1 - q^5} - \dots = 0. \end{aligned}$$

The second and third lines unite together, and the equation becomes

$$\begin{aligned} &-\frac{1}{1 + q} + \frac{2q}{1 + q^2} - \frac{3q^2}{1 + q^3} + \frac{4q^3}{1 + q^4} - \dots \\ &+ \frac{2}{1 - q} - \frac{4q}{1 + q^2} + \frac{6q^2}{1 - q^3} - \frac{8q^3}{1 + q^4} + \dots \\ &-\frac{1}{1 - q} - \frac{3q^2}{1 - q^3} - \frac{5q^4}{1 - q^5} - \frac{7q^6}{1 - q^7} - \dots = 0; \end{aligned}$$

or, collecting and arranging,

$$\begin{aligned} & -\frac{1}{1+q} - \frac{2q}{1+q^2} - \frac{3q^2}{1+q^3} - \frac{4q^3}{1+q^4} - \frac{5q^4}{1+q^5} - \dots \\ & + \frac{1}{1-q} + \frac{3q^2}{1-q^3} + \frac{5q^4}{1-q^5} + \dots = 0, \end{aligned}$$

an identity which it is easy to verify to any number of terms. But to prove it directly, we have only to add the pairs of terms in the alternate columns; calling the left-hand side Fq , we thus obtain

$$\begin{aligned} Fq = 2q \left\{ -\frac{1}{1+q^2} - \frac{2q^2}{1+q^4} - \frac{3q^4}{1+q^6} - \dots \right. \\ \left. + \frac{1}{1-q^2} + \frac{3q^4}{1-q^6} + \dots \right\}; \end{aligned}$$

viz. this equation is $Fq = 2qF(q^2)$; and thence

$$Fq = 2^2 q^{1+2} F(q^4) = 2^3 q^{1+2+4} F(q^8) = \&c.;$$

we thus have $Fq = 0$.

The equation (B), or, what is the same thing, the equation (A) is thus proved.

Reverting to the equation (A), we have

$$(1 + 2q + 2q^4 + \dots)^4 = \frac{4K^2}{\pi^2},$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. I., p. 239),

$$\left(\frac{q}{1-q^2} + \frac{2q^2}{1-q^4} + \dots \right) = \frac{K^2}{2\pi^2} \left(1 - \frac{E_1}{K} \right),$$

(ib., p. 135; ib., p. 189),

if $q = e^{-\frac{\pi K'}{K}}$, and K, E_1 are the complete functions $F_1 k, E_1 k$.

The left-hand side of the equation is thus

$$\frac{4K^2}{\pi^2} - \frac{8K^2}{\pi^2} \left(1 - \frac{E_1}{K} \right) = \frac{4K^2}{\pi^2} \left(-1 + \frac{2E_1}{K} \right),$$

and we have

$$\left(-1 + \frac{2E_1}{K} \right) = \frac{\pi^2}{4K^2} \cdot \frac{1 - 9q^4 - 25q^8 + 49q^{12} + 81q^{16} - \dots}{1 - q^4 - q^8 + q^{12} + q^{16} - \dots},$$

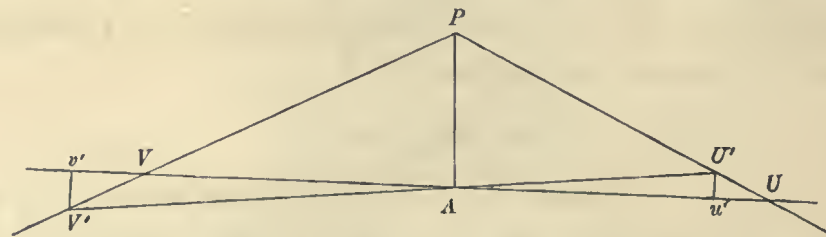
which is a new expression for E_1 as a q -function. The expression on the right-hand side presents itself, Clebsch, *Theorie der Elasticität* (Leipzig, 1862), p. 162, and must have been obtained by him as a value for $\left(-1 + \frac{2E_1}{K} \right)$; but there is no statement that this is so, nor anything to show how this form of q -function was arrived at. Mr Todhunter called my attention to the passage in Clebsch.

639.

AN ELEMENTARY CONSTRUCTION IN OPTICS.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 81, 82.]

CONSIDER two lines meeting at a point P , and a point A ; through A , draw at right angles to AP , a line meeting the two lines in the points U , V respectively; and through the same point A draw any other line meeting the two lines in the



points U' , V' respectively; also let the points u' , v' be the feet of the perpendiculars let fall from U' , V' respectively on the line UV ; then we have

$$\frac{1}{Au'} + \frac{1}{Av'} = \frac{1}{AU} + \frac{1}{AV}.$$

The theorem can be proved at once without any difficulty. It answers to the optical construction, according to which, if UPV represents the path of a ray through a convex lens AP , then the thin pencil, axis $U'P$ and centre U' , converges after refraction to the point V' , where $U'V'$ are *in line* with A the centre of the lens; considering as usual the inclinations to the axis as small, we have approximately $AV' = Av'$, $AU' = Au'$, and the theorem is

$$\frac{1}{AU'} + \frac{1}{AV'} = \frac{1}{AU} + \frac{1}{AV}, = \frac{1}{AF},$$

if AF is the focal length of the lens.

In the original theorem, the line UV need not be at right angles to AP , but may be any line whatever; the projecting lines $U'u'$ and $V'v'$ must then be parallel to AP , and the theorem remains true.

640.

FURTHER NOTE ON MR MARTIN'S PAPER "ON THE INTEGRALS
OF SOME DIFFERENTIALS."

[From the *Messenger of Mathematics*, vol. VI. (1877), p. 82.]

See paper, Number 633; this further note relates also to a detail.

641.

ON THE FLEXURE OF A SPHERICAL SURFACE.

[From the *Messenger of Mathematics*, vol. VI. (1877), pp. 88—90.]

It is known that an inextensible spherical surface, or to fix the ideas the spherical quadrilateral included between two arcs of meridian and two arcs of parallel, may be bent in suchwise as to be part of a surface of revolution, the meridians and parallels of the spherical surface being meridians and parallels of the new surface, and, moreover, the radius of each parallel of the spherical surface being in the new surface altered in the constant ratio k to 1. We have, in fact, on the spherical surface, writing p for the latitude and q for the longitude, and the radius being unity,

$$x = \cos p \cos q,$$

$$y = \cos p \sin q,$$

$$z = \sin p,$$

values which give

$$dx^2 + dy^2 + dz^2 = dp^2 + \cos^2 p \, dq^2.$$

This last equation is satisfied by the values

$$x = \cos p \cos \frac{q}{k},$$

$$y = \cos p \sin \frac{q}{k},$$

$$z = E(k, p),$$

where $E(k, p) = \int_0^p \sqrt{1 - k^2 \sin^2 p} \, dp$, is the elliptic function of the second kind; or rather, this is so when $k < 1$, but the same notation may be used when $k > 1$. These values give the deformation in question.

The two cases to be considered are $k < 1$, and $k > 1$; we take in each case a spherical quadrilateral $ABCD$ (fig. 1), bounded by AB (an arc of the equator), the arc of parallel CD , and the two arcs of meridian AD and BC . In the first case, there is no limit to the latitude $AD, = BC$, or taking these $= 90^\circ$, we may in place